



Baldivis
Secondary College

Name _____

Year 8

Class 8- _____

Investigation 2

Teacher: _____

Statistics

Time: 40 min

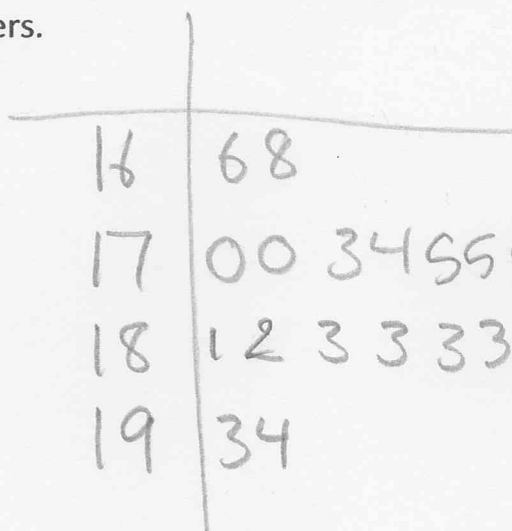
Marks: ____ /

31

"Hey Diddle diddle the median's the middle,
You add and divide for the mean.
The mode is the one that appears the most,
And the range is the difference between."

Heights of Australian Prime Ministers (cm)
166, 193, 184, 183, 182, 181, 176, <u>178</u> , 173, 170, 175, 175, 168, 183, 183, 179, 174, 170, 175, 178, 194, <u>178</u> , 183, <u>178</u>

1. Create a stem and leaf plot represent the heights of Australian Prime Ministers.

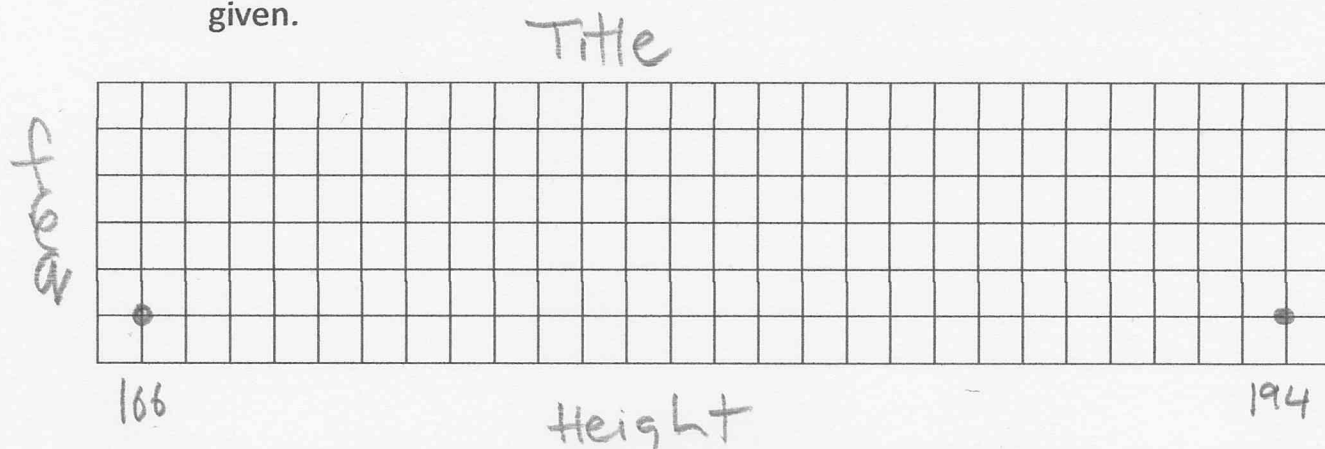


Ascending
Order.

2. Find the mean, mode, median and range for the heights of Australian Prime Ministers.

	Working out	Answer
MEAN	$4279 \div 24$	178.29 ✓
MEDIAN	13 th NUMBER.	178 ✓
MODE	BI MODAL	178/183 ✓
RANGE	$194 - 166 =$	28 ✓

3. Create a dot plot to visually represent the information you have been given.



4. What information can you extract from the data you have been given?

Any 3 facts.

5. What would you expect to happen to the mean, mode, median and range if you remove the 3 lowest heights?

	Original	New	Explanation for change
MEAN	178.29	179.76	Removing Lowest $\uparrow$ mean...
MEDIAN	178	178	No change
MODE	178/183	178/183	No change
RANGE	28	24	Shorter, Spread

6. What new height can be added to the original data set to create a new median?

No new height.

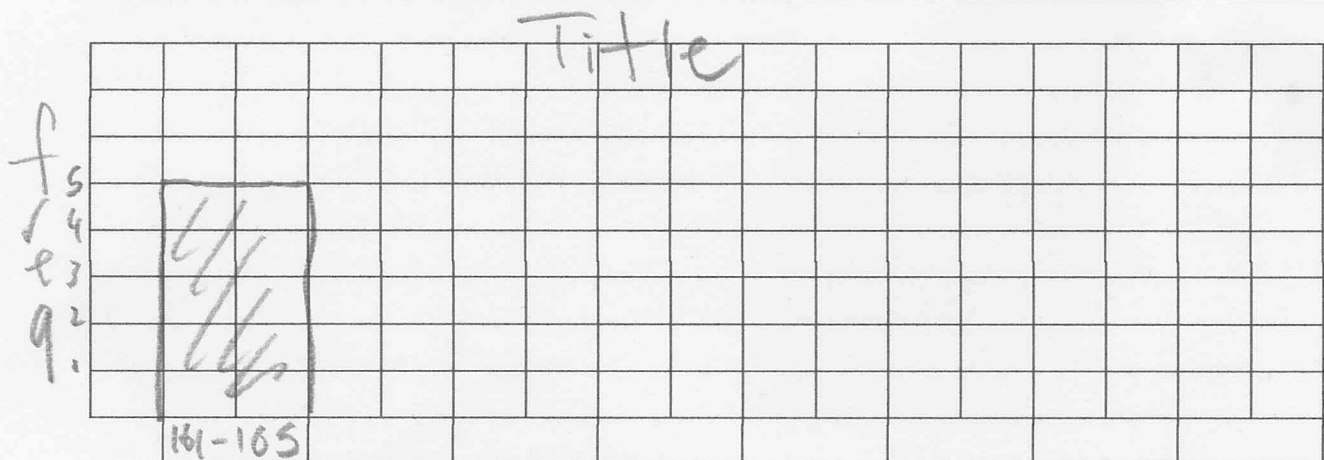
7. What would happen to the mean if you changed the mode to 166cm by adding extra heights? (Show working out)

$$+ (166 \times 4)$$

$$4943 \div 28 = 176.54$$

8. Create a frequency column graph to visually represent this new set of data.

161-165	166-170	171-175	176-180	181-185	186-190	191-195	196-200



9. What effect does removing the tallest Prime Minister have on the

	Original	New	Explanation for result
MEAN	178.29	177.6	↓
MEDIAN	178	178	No change
MODE	178/183	178/183	No change
RANGE	28	27	Smaller Spread.

End of investigation